Final Experimental Protocol

PDF edition of preserved repository document | docs/final_experimental_protocol.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Frozen final protocol

The historical 8722-parameter MLP is retained. Final primary conditions use the same frozen Apache-2.0 SmolLM2-135M-Instruct transformer and the same two-layer task-head architecture, initialization seed and nested training problem IDs. Pretrained weights are not fine-tuned: this is a frozen-feature adaptation study, not an instruction-tuned arithmetic model or a claim that a language model is intrinsically stronger at this task. Its actual results will determine quality.

Canonical model inputs are numerical operands for answer-only, operands with goal/length markers for structured, and current cursor/local operand digits/partial digit/carry/position/phase for procedural B/C. No product, expected claim, reference future state or verifier label enters inference. The environment retains hand-defined long multiplication sequencing; the learner chooses among 100 numerical claims. This domain scaffold is a limitation, not discovery of arbitrary procedures. The new head removes seven dummy policy losses and the artificial skip/sign score coupling. The output capacity is identical; only active supervision heads differ by task.

A new seed-5151 dataset contains 1000 training, 100 validation, 200 IID and 300 structural OOD examples. Structures remain train 1x1/2x1/2x2 and OOD 3x2/3x3/4x2. No test-based hyperparameter selection. Budgets 10/25/50/100/250/500/1000; seeds 0/1/2. Fixed 1200 Adam updates, batch64, lr0.003; final checkpoint. Data beyond1000 is deferred to avoid expanding scope to chase targets. Validation is diagnostic, never selection. All original test sets remain historical.

B receives teacher transition labels. C receives only its own accepted proposals and bounded scalar feedback (64 attempts/problem), followed by replay. A/structured receive final-answer labels. A matched-information control trains the same procedural backbone/head on B's exact local state/action rows without episodic or dream components; it is a conventional local supervised policy, not an answer-only learner receiving hidden labels. It establishes whether the wrapper adds anything beyond the representation/supervision. Costs report rows, consumed minibatch rows, optimizer steps, transformer input tokens/model calls, scalar feedback attempts, time and shared extraction cost separately.

Primary dream is disabled to avoid confidence lookahead cost; a bounded depth2/branch3 intervention measures it at n=50. No-state retrains B at n=50; no-memory matches C's accepted samples and optimizer update count using in-memory arrays instead of SQLite replay. No interaction performs proposed-state recurrence, with hypothetical evidence only. Goal evaluator removal is a diagnostic scoring removal, not a changed arithmetic test; no goal oracle exists in the policy. No-process-supervision uses the C regime and is explicitly a supervision/query contrast. All primary seeds are retained. Single-budget ablations cannot estimate delta N*.

Shared evaluator reports exact answer, verified trajectory, terminal goal and groundedness separately. Baseline trajectories are absent and validity is null. Groundedness comparisons inherit this output-protocol difference. N* uses observed mean-accuracy thresholds 0.8/0.9/0.95, no interpolation; missing crossings and DER are not_reached. Mean±sample SD, per-run Wilson95 intervals and paired exploratory McNemar at

n=1000 seed 0 with Holm correction; no seed pooling or claim from significance alone. Errors group by condition/budget/split/digit structure/carry count. Only actual frozen runs enter results/final.

The checkpoint revision is 12fd25f77366fa6b3b4b768ec3050bf629380bac. Actual frozen backbone134,515,008 parameters; trainable head337,618; total134,852,626. CPU execution because the installed runtime cannot access MPS. The categorical numeric feature adapter performs indexing/quantization only; it never multiplies digits or computes labels. The text encoder sees indexed numerical features, not natural instruction prompts. This limits interpretation of language/math pretraining benefits. All conditions share this adapter and head capacity.

No-goal-evaluator intervention removes the constant model-visible goal flag; external scoring remains unchanged. It tests that feature's contribution, not removal of benchmark correctness. The matched-information policy bypasses working/concept memory while preserving the exact learned predictor, local inputs and real execution evidence. No-memory arrays are in the same chronological order as episodic replay and use exactly1200 updates: it tests storage/retrieval, not the benefit of reuse itself. Cache reuse is shared and target-free; warm/cached latency is not presented as cold deployment latency or compute superiority. Tokens actually processed by the frozen transformer and feature requests are reported separately from gradient-update rows.